Student name: Nguyen Bao Chinh

Student ID: 24520225

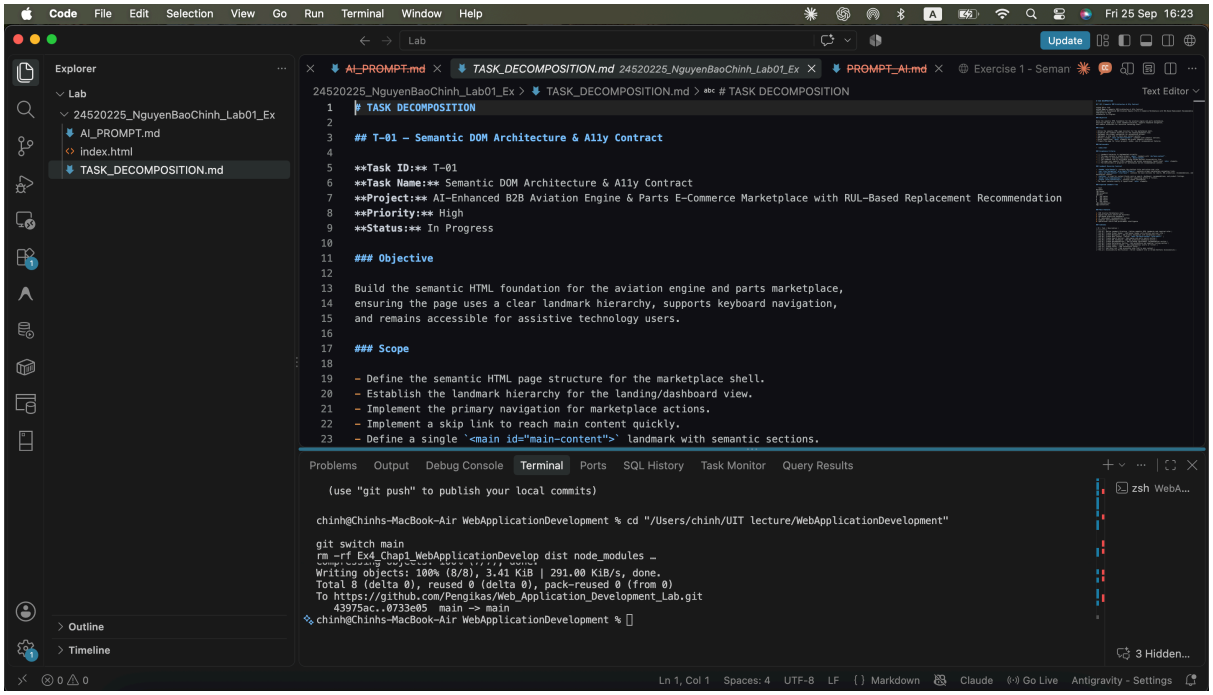
Link Github: https://github.com/Pengikas/Web_Application_Development_Lab.git

EXERCISE 1

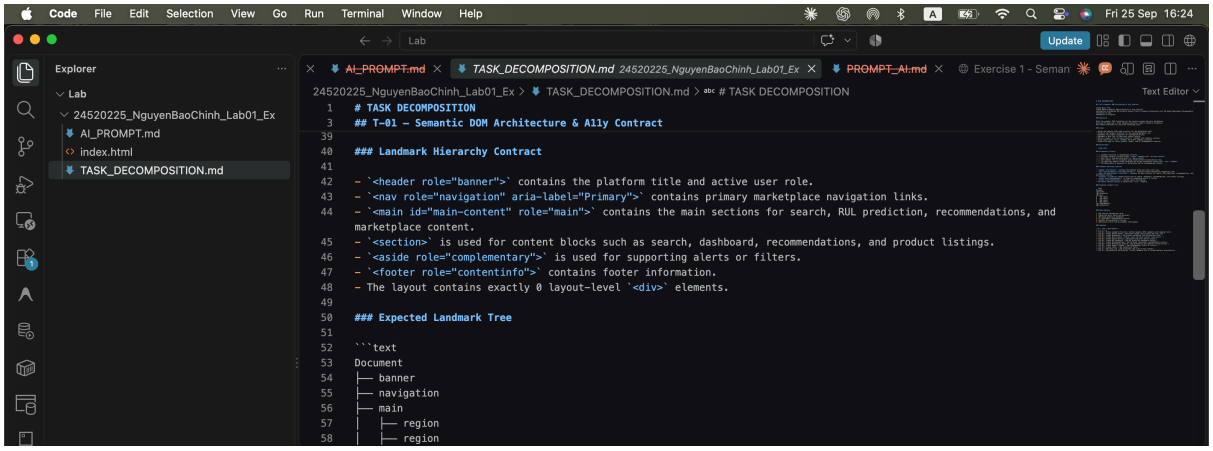
Link AI Agent: https://claude.ai/share/f16d5926-93a6-46bd-b3bd-9ceaf7d9896d

Screenshot for the results

Step 1: WBS Task T-01 in TASK_DECOMPOSITION.md.



Step 2: Define landmark hierarchy contract (0 elements).



Expected Landmark Tree

```text
Document
├── banner
├── navigation
├── main
│   ├── region
│   ├── region
│   ├── region
│   └── region
├── complementary
└── contentinfo
```

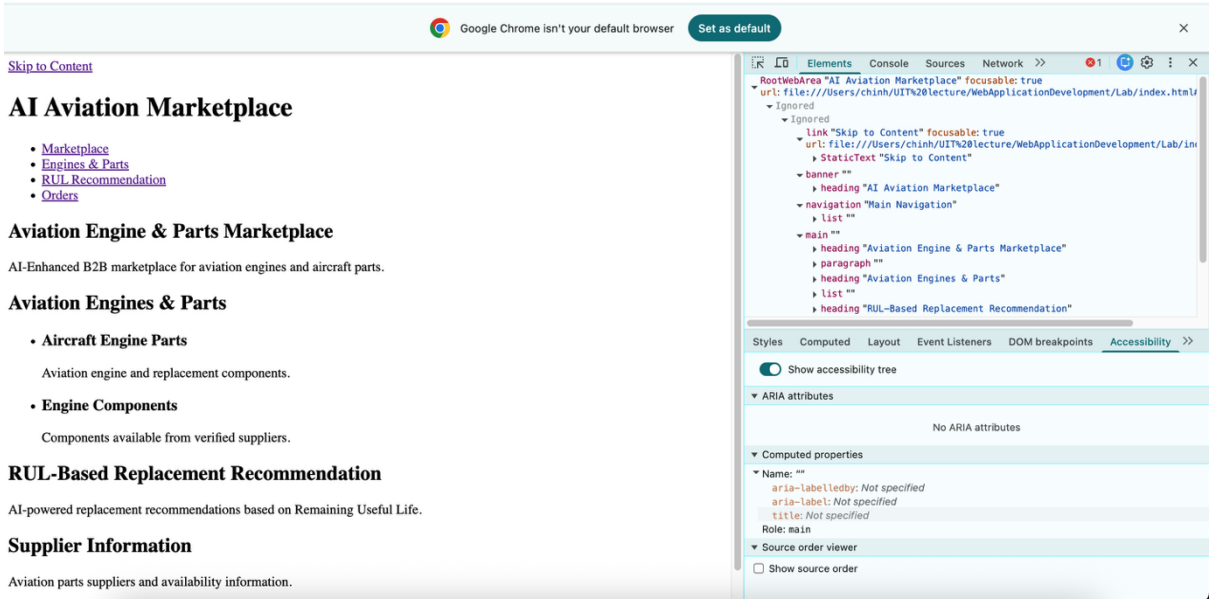
Step 3: Implement accessible skip-link (Skip to Content).



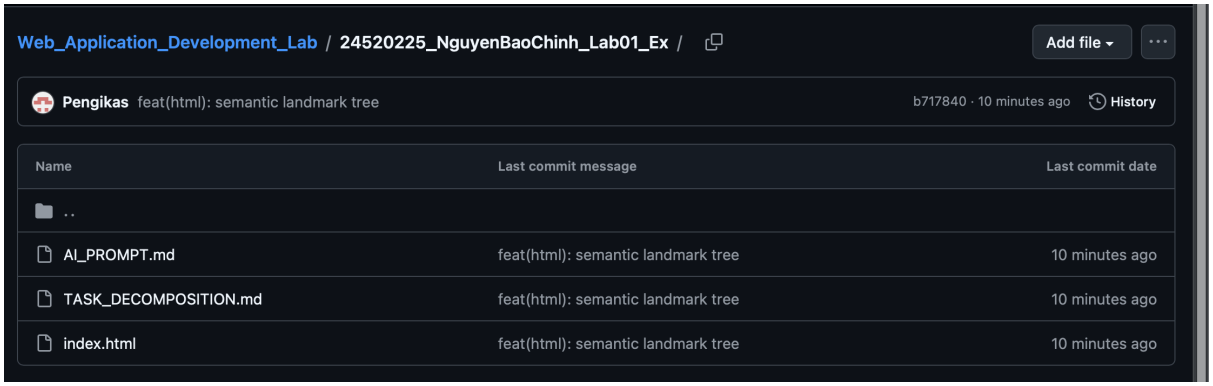
Step 4: Atomic Commit: git commit -m 'feat(html): semantic landmark tree'



- **Open Chrome DevTools -> Accessibility -> Verify Landmark Tree.**



One-shot prompt penalty: Commits combining CSS with HTML get 0 pts.

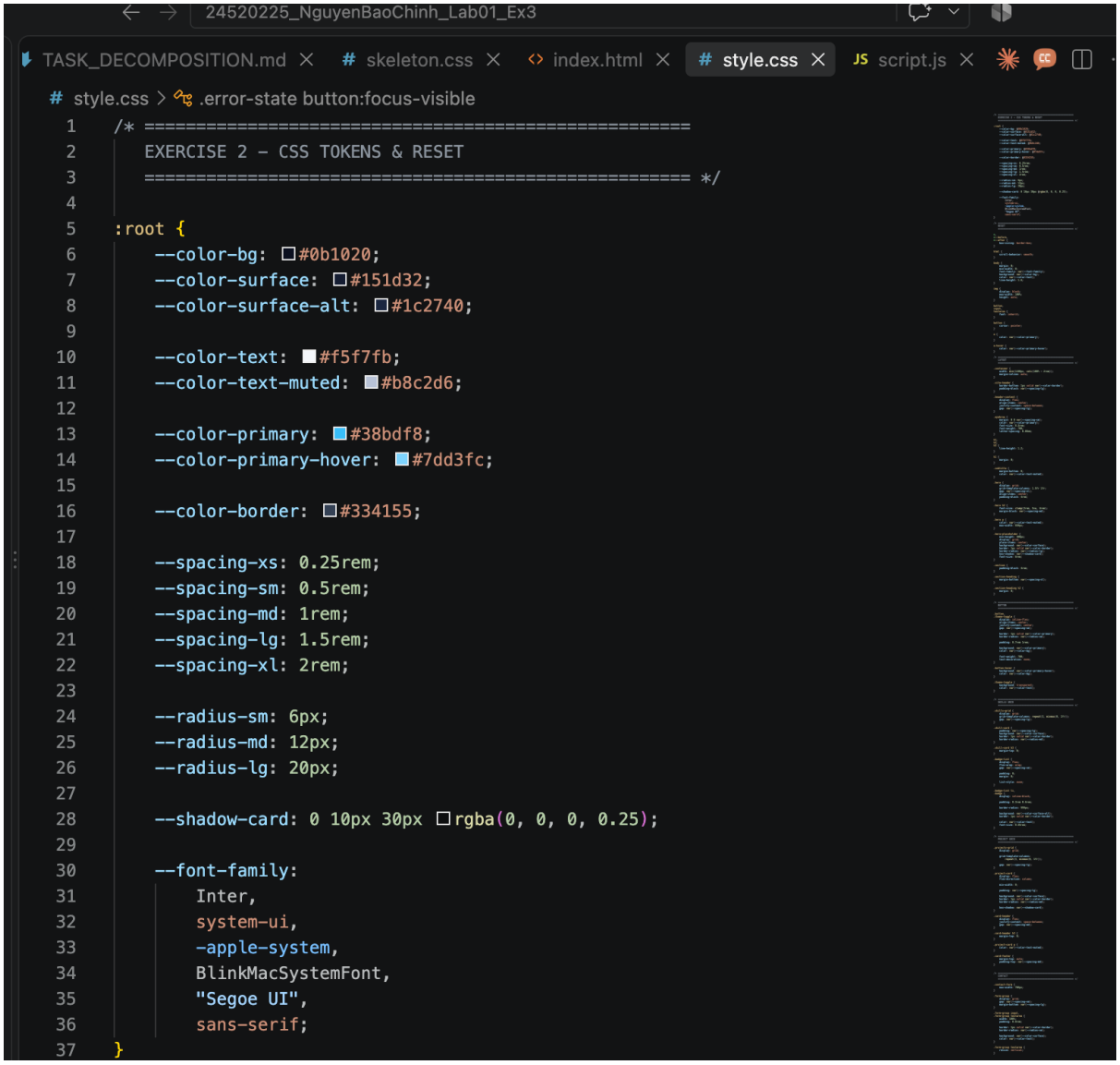


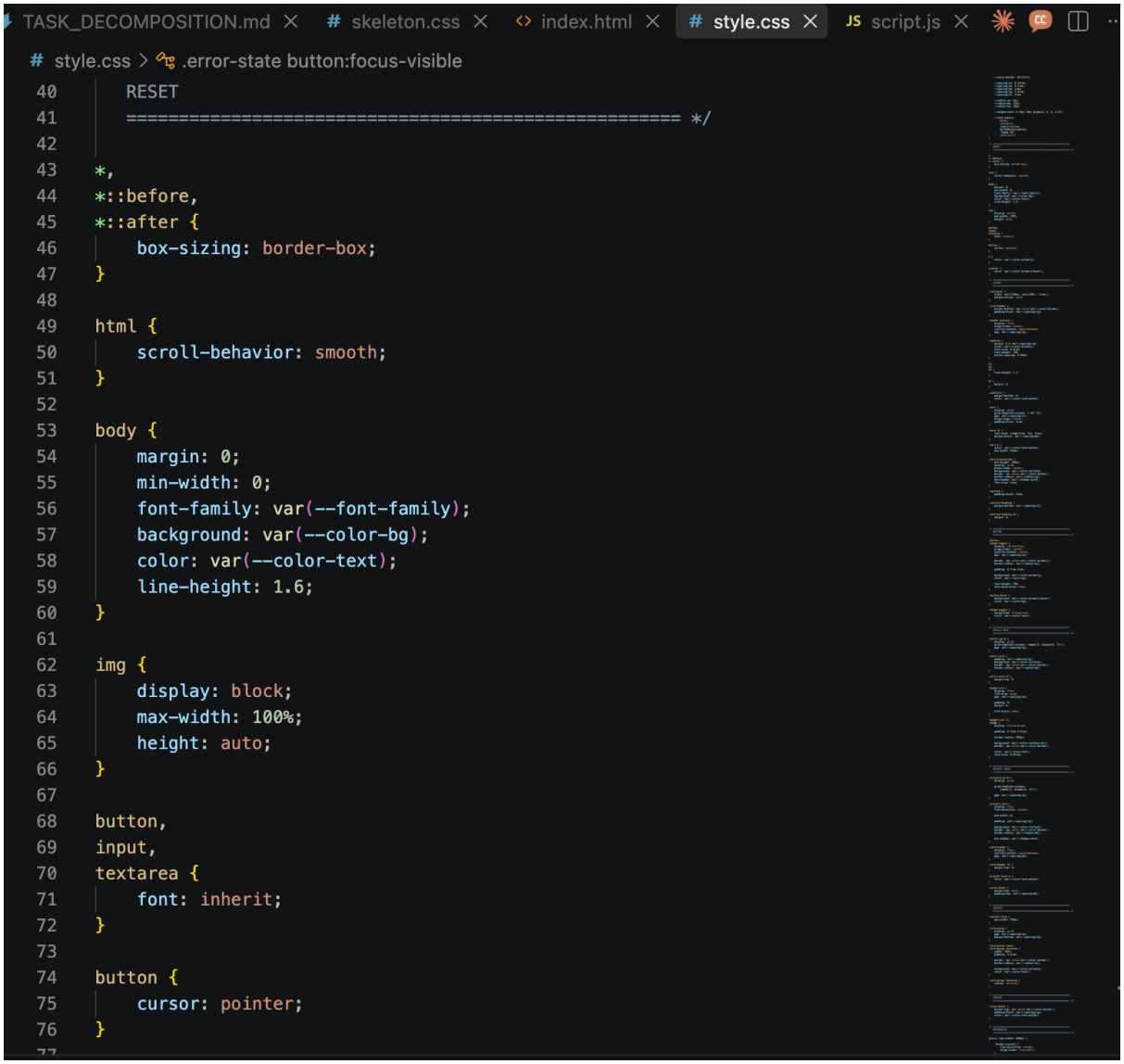
EXERCISE 2

Link AI prompt: https://chatgpt.com/share/6ab6a30b-9340-83ec-af80-1db60ee8609d?ogimg=plain

Screenshot for results

- **SUB-TASK T-02A: Tokens & Reset -> git commit -m 'feat(css): tokens & reset'**


24520225_NguyenBaoChinh_Lab01_Ex3
TASK_DECOMPOSITION.md
skeleton.css
index.html
style.css
script.js
style.css > .error-state button:focus-visible
/* ==========
EXERCISE 2 - CSS TOKENS & RESET
========== */
:root {
--color-bg: #0b1020;
--color-surface: #151d32;
--color-surface-alt: #1c2740;
--color-text: #f5f7fb;
--color-text-muted: #b8c2d6;
--color-primary: #38bdf8;
--color-primary-hover: #7dd3fc;
--color-border: #334155;
--spacing-xs: 0.25rem;
--spacing-sm: 0.5rem;
--spacing-md: 1rem;
--spacing-lg: 1.5rem;
--spacing-xl: 2rem;
--radius-sm: 6px;
--radius-md: 12px;
--radius-lg: 20px;
--shadow-card: 0 10px 30px rgba(0, 0, 0, 0.25);
--font-family:
Inter,
system-ui,
-apple-system,
BlinkMacSystemFont,
"Segoe UI",
sans-serif;
}


TASK_DECOMPOSITION.md
skeleton.css
index.html
style.css
script.js
style.css > .error-state button:focus-visible
RESET
*/
*,
*::before,
*::after {
box-sizing: border-box;
}
html {
scroll-behavior: smooth;
}
body {
margin: 0;
min-width: 0;
font-family: var(--font-family);
background: var(--color-bg);
color: var(--color-text);
line-height: 1.6;
}
img {
display: block;
max-width: 100%;
height: auto;
}
button,
input,
textarea {
font: inherit;
}
button {
cursor: pointer;
}

```css
html {
    scroll-behavior: smooth;
}

body {
    margin: 0;
    min-width: 0;
    font-family: var(--font-family);
    background: var(--color-bg);
    color: var(--color-text);
    line-height: 1.6;
}

img {
    display: block;
    max-width: 100%;
    height: auto;
}

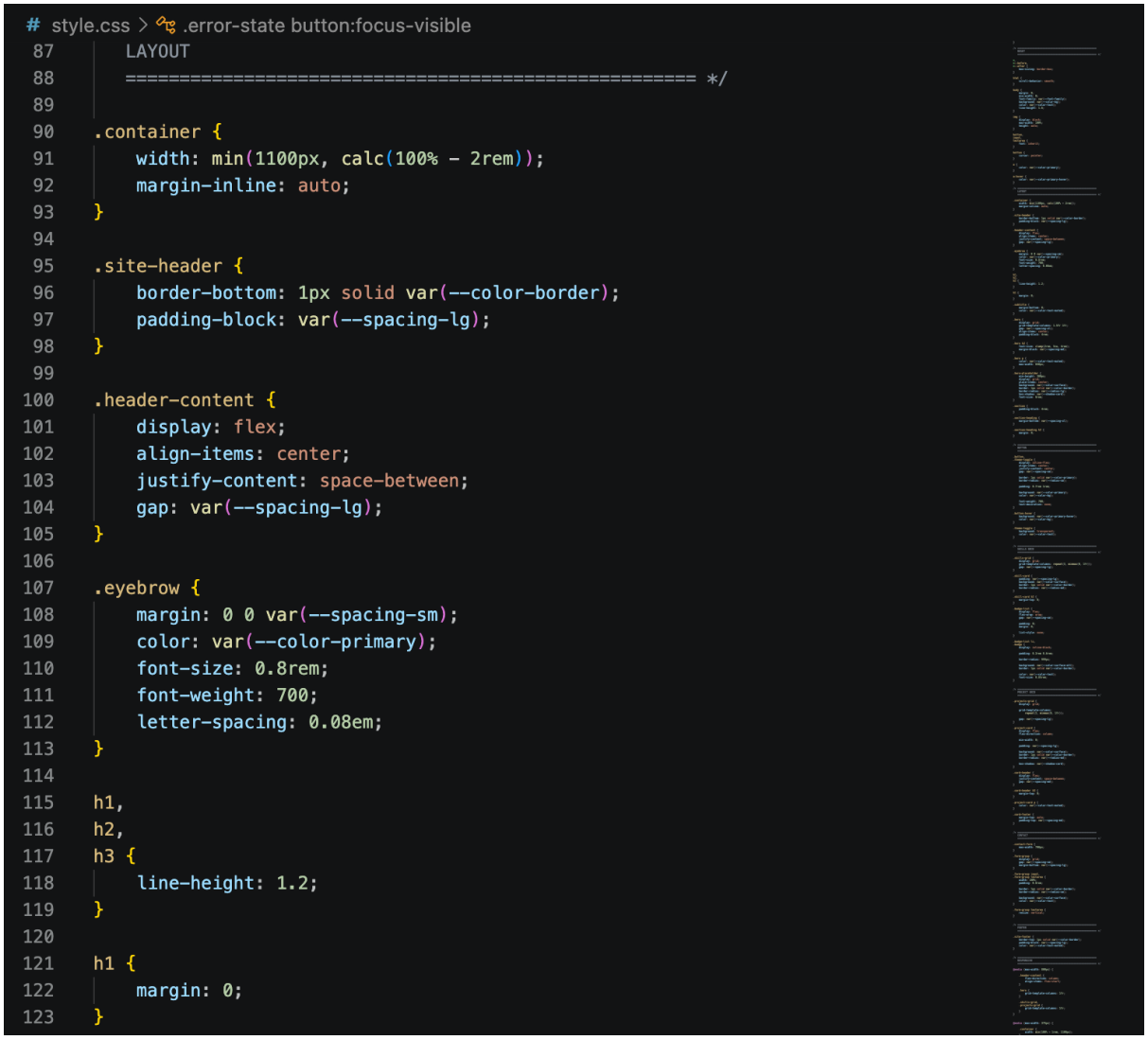
button,
input,
textarea {
    font: inherit;
}

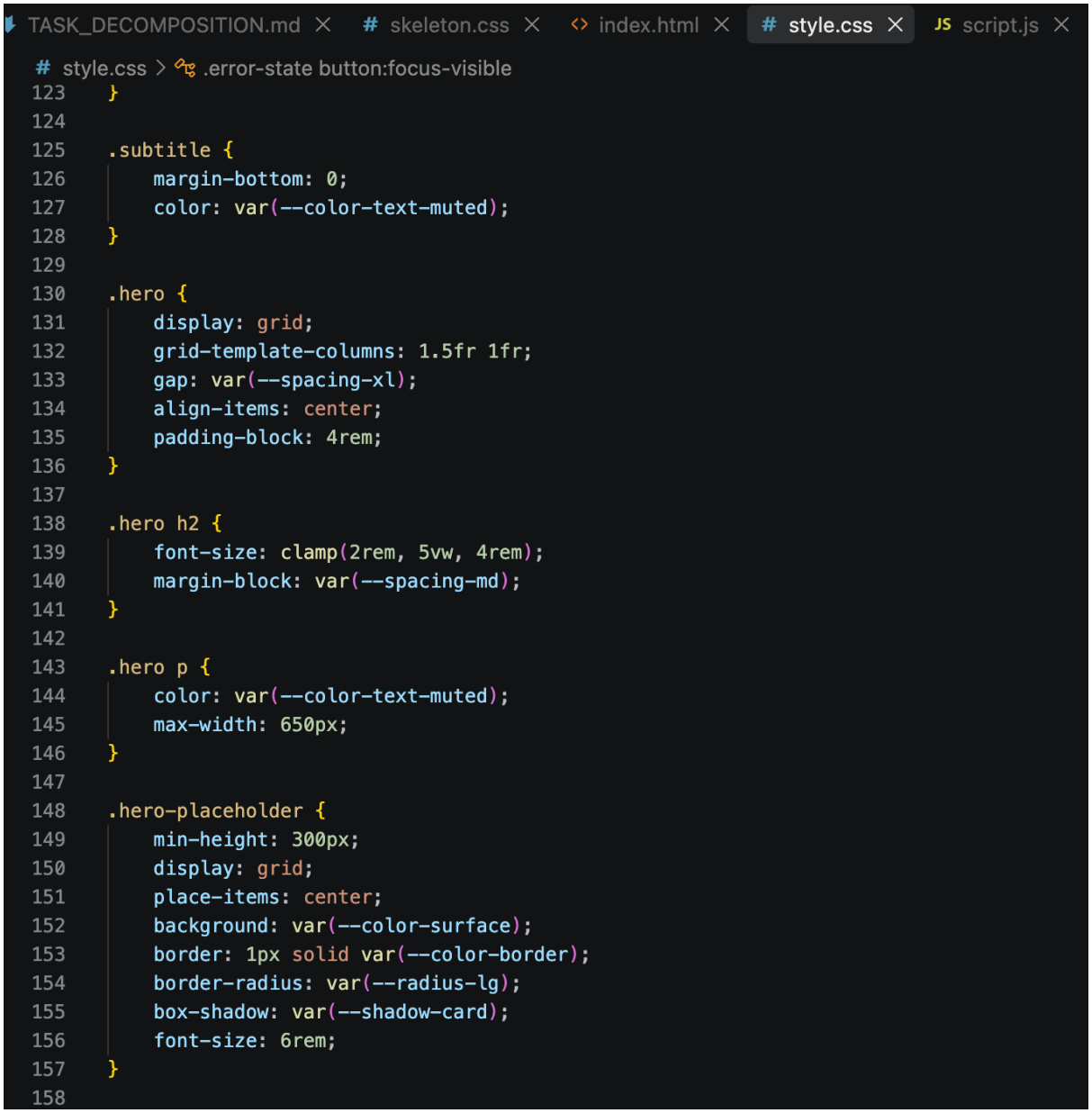
button {
    cursor: pointer;
}

a {
    color: var(--color-primary);
}

a:hover {
    color: var(--color-primary-hover);
}
```

- **SUB-TASK T-02B: 2D Grid Layout -> git commit -m 'feat(css): responsive grid'**


style.css > .error-state button:focus-visible
LAYOUT
========== */
.container {
width: min(1100px, calc(100% - 2rem));
margin-inline: auto;
}
.site-header {
border-bottom: 1px solid var(--color-border);
padding-block: var(--spacing-lg);
}
.header-content {
display: flex;
align-items: center;
justify-content: space-between;
gap: var(--spacing-lg);
}
.eyebrow {
margin: 0 0 var(--spacing-sm);
color: var(--color-primary);
font-size: 0.8rem;
font-weight: 700;
letter-spacing: 0.08em;
}
h1,
h2,
h3 {
line-height: 1.2;
}
h1 {
margin: 0;
}


TASK_DECOMPOSITION.md
skeleton.css
index.html
style.css
script.js
style.css > .error-state button:focus-visible
123 }
124
125 .subtitle {
126 margin-bottom: 0;
127 color: var(--color-text-muted);
128 }
129
130 .hero {
131 display: grid;
132 grid-template-columns: 1.5fr 1fr;
133 gap: var(--spacing-xl);
134 align-items: center;
135 padding-block: 4rem;
136 }
137
138 .hero h2 {
139 font-size: clamp(2rem, 5vw, 4rem);
140 margin-block: var(--spacing-md);
141 }
142
143 .hero p {
144 color: var(--color-text-muted);
145 max-width: 650px;
146 }
147
148 .hero-placeholder {
149 min-height: 300px;
150 display: grid;
151 place-items: center;
152 background: var(--color-surface);
153 border: 1px solid var(--color-border);
154 border-radius: var(--radius-lg);
155 box-shadow: var(--shadow-card);
156 font-size: 6rem;
157 }
158



```
}

.section {
    padding-block: 4rem;
}

.section-heading {
    margin-bottom: var(--spacing-xl);
}

.section-heading h2 {
    margin: 0;
}


/* ========================================================
   BUTTON
   ======================================================== */

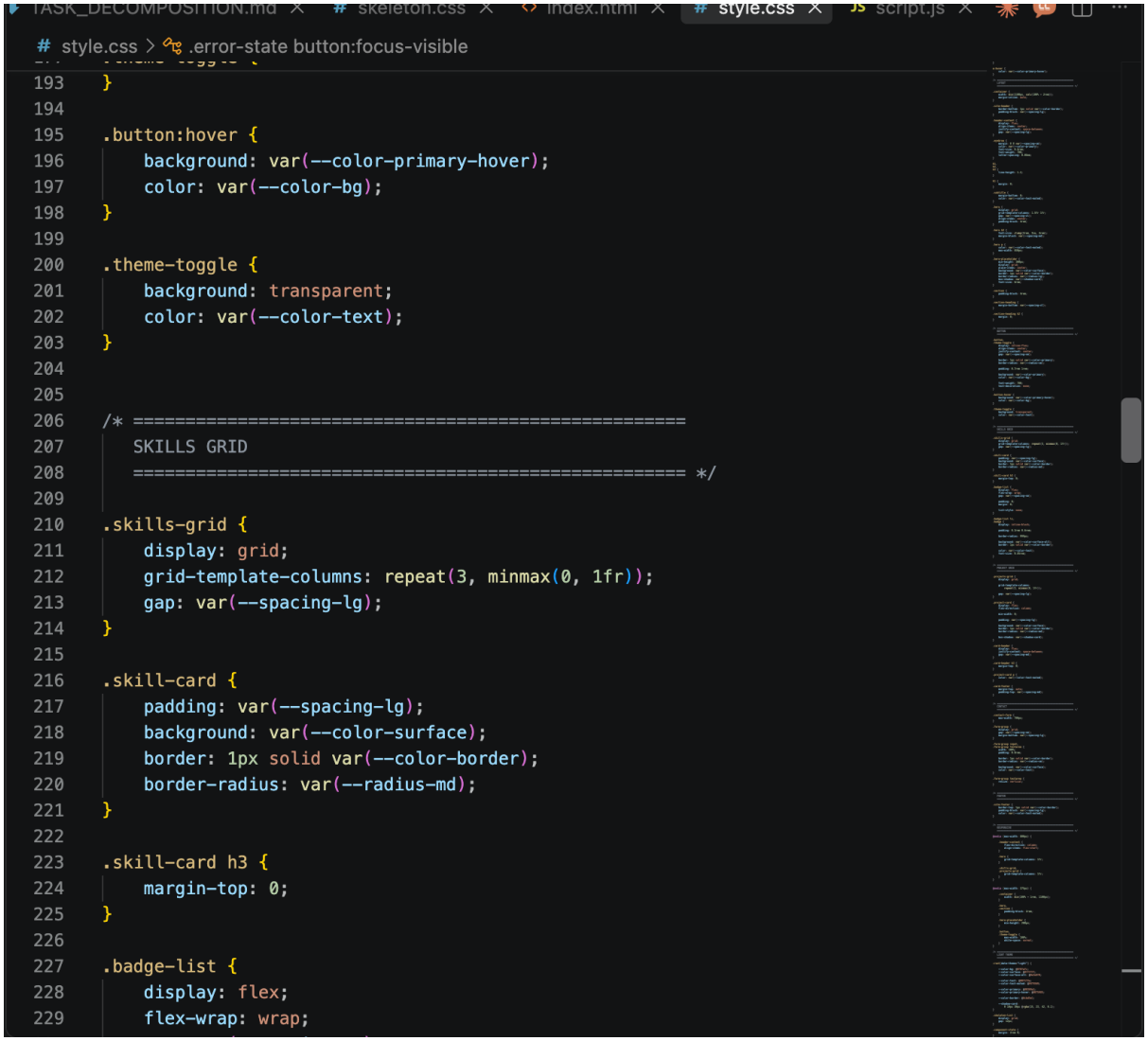
.button,
.theme-toggle {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: var(--spacing-sm);

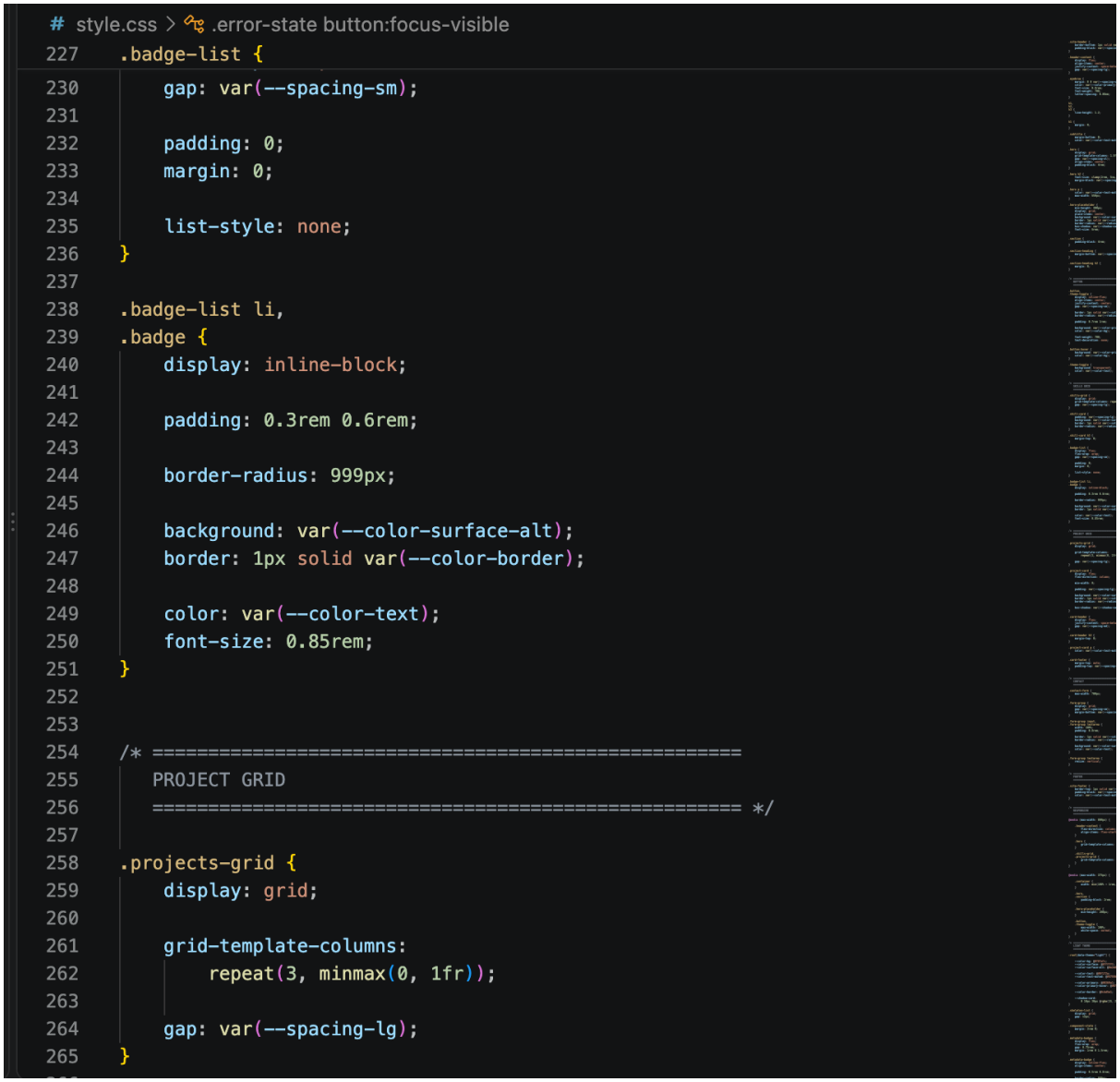
    border: 1px solid var(--color-primary);
    border-radius: var(--radius-sm);

    padding: 0.7rem 1rem;

    background: var(--color-primary);
    color: var(--color-bg);

    font-weight: 700;
    text-decoration: none;
}
```


TASK_DECOMPOSITION.md
skeleton.css
index.html
style.css
script.js
style.css > .error-state button:focus-visible
193 }
194
195 .button:hover {
196 background: var(--color-primary-hover);
197 color: var(--color-bg);
198 }
199
200 .theme-toggle {
201 background: transparent;
202 color: var(--color-text);
203 }
204
205
206 /* ==
207 SKILLS GRID
208 == */
209
210 .skills-grid {
211 display: grid;
212 grid-template-columns: repeat(3, minmax(0, 1fr));
213 gap: var(--spacing-lg);
214 }
215
216 .skill-card {
217 padding: var(--spacing-lg);
218 background: var(--color-surface);
219 border: 1px solid var(--color-border);
220 border-radius: var(--radius-md);
221 }
222
223 .skill-card h3 {
224 margin-top: 0;
225 }
226
227 .badge-list {
228 display: flex;
229 flex-wrap: wrap;


style.css > .error-state button:focus-visible
227 .badge-list {
230 gap: var(--spacing-sm);
231
232 padding: 0;
233 margin: 0;
234
235 list-style: none;
236 }
237
238 .badge-list li,
239 .badge {
240 display: inline-block;
241
242 padding: 0.3rem 0.6rem;
243
244 border-radius: 999px;
245
246 background: var(--color-surface-alt);
247 border: 1px solid var(--color-border);
248
249 color: var(--color-text);
250 font-size: 0.85rem;
251 }
252
253
254 /* ==
255 PROJECT GRID
256 == */
257
258 .projects-grid {
259 display: grid;
260
261 grid-template-columns:
262 repeat(3, minmax(0, 1fr));
263
264 gap: var(--spacing-lg);
265 }

```
301
302 /* ========================================================
303    CONTACT
304    ======================================================== */
305
306 .contact-form {
307     max-width: 700px;
308 }
309
310 .form-group {
311     display: grid;
312     gap: var(--spacing-sm);
313     margin-bottom: var(--spacing-lg);
314 }
315
316 .form-group input,
317 .form-group textarea {
318     width: 100%;
319     padding: 0.8rem;
320
321     border: 1px solid var(--color-border);
322     border-radius: var(--radius-sm);
323
324     background: var(--color-surface);
325     color: var(--color-text);
326 }
327
328 .form-group textarea {
```

```
/* ======================================================
   FOOTER
   ====================================================== */

.site-footer {
    border-top: 1px solid var(--color-border);
    padding-block: var(--spacing-lg);
    color: var(--color-text-muted);
}


/* ======================================================
   RESPONSIVE
   ====================================================== */

@media (max-width: 800px) {

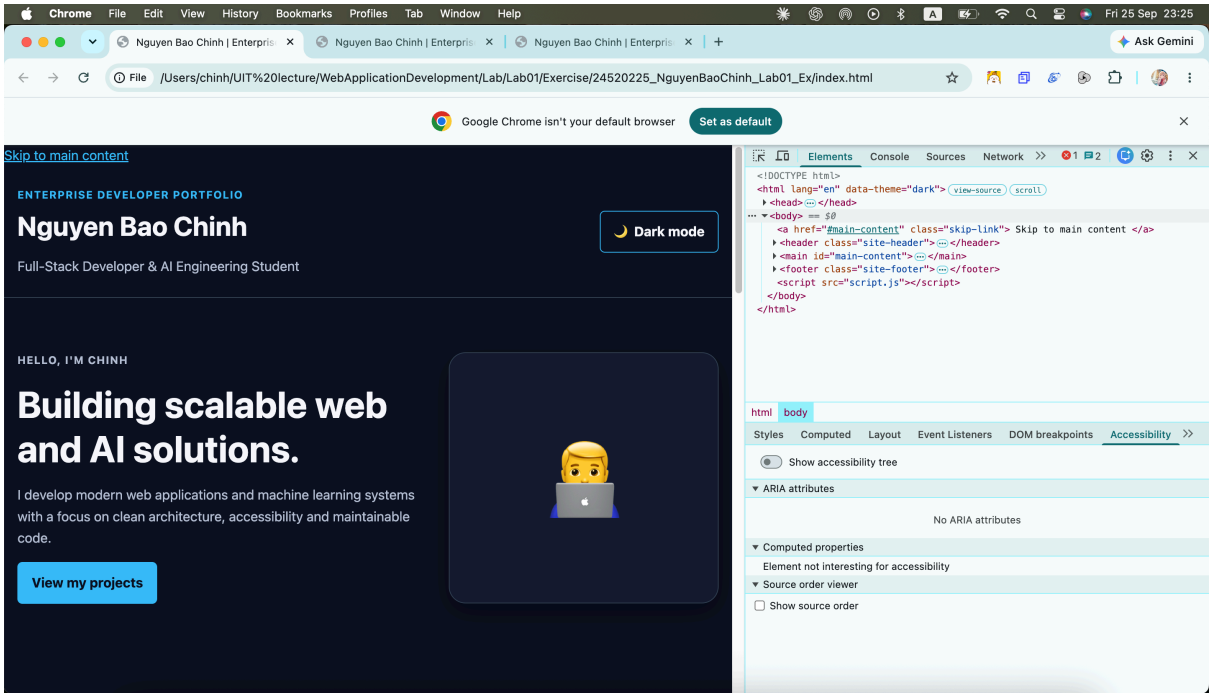
    .header-content {
        flex-direction: column;
        align-items: flex-start;
    }

    .hero {
        grid-template-columns: 1fr;
    }

    .skills-grid,
    .projects-grid {
        grid-template-columns: 1fr;
    }
}

```

- **SUB-TASK T-02C: Theme Engine -> git commit -m 'feat(js): dark mode engine'**


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Fri 25 Sep 23:25
Nguyen Bao Chinh | Enterpris
Ask Gemini
File /Users/chinh/UIT%20lecture/WebApplicationDevelopment/Lab/Lab01/Exercise/24520225_NguyenBaoChinh_Lab01_Ex/index.html
Google Chrome isn't your default browser
Set as default
Skip to main content
ENTERPRISE DEVELOPER PORTFOLIO
Nguyen Bao Chinh
Full-Stack Developer & AI Engineering Student
Dark mode
HELLO, I'M CHINH
Building scalable web and AI solutions.
I develop modern web applications and machine learning systems with a focus on clean architecture, accessibility and maintainable code.
View my projects
Elements Console Sources Network
<!DOCTYPE html>
<html lang="en" data-theme="dark">
<head>…</head>
<body> == $0
Skip to main content
<header class="site-header">…</header>
<main id="main-content">…</main>
<footer class="site-footer">…</footer>
<script src="script.js"></script>
</body>
</html>
html body
Styles Computed Layout Event Listeners DOM breakpoints Accessibility
Show accessibility tree
ARIA attributes
No ARIA attributes
Computed properties
Element not interesting for accessibility
Source order viewer
Show source order

TASK_DECOMPOSITION.md | skeleton.css | index.html | style.css | script.js

script.js > ...

```js
const themeToggle = document.querySelector("#theme-toggle");
const themeIcon = document.querySelector("#theme-icon");
const themeText = document.querySelector("#theme-text");
const contactForm = document.querySelector("#contact-form");

const savedTheme = localStorage.getItem("theme");

function applyTheme(theme) {
    const isDark = theme === "dark";

    document.documentElement.dataset.theme = theme;

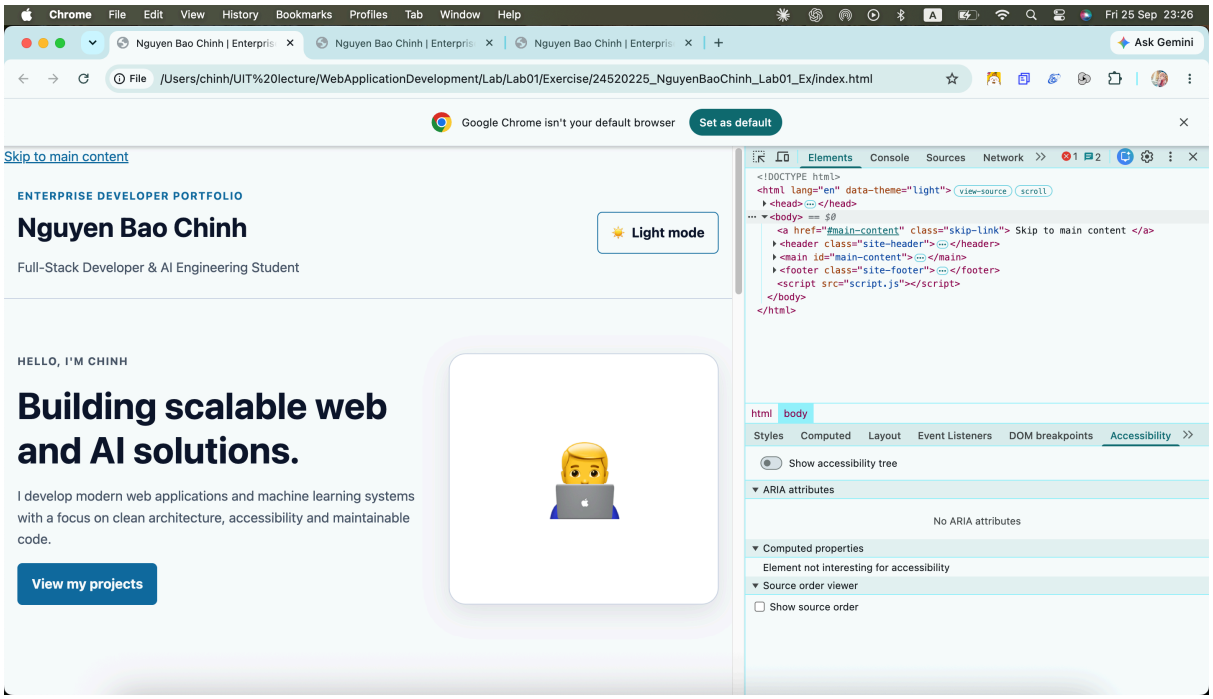
    themeToggle.setAttribute(
        "aria-pressed",
        String(isDark)
    );

    themeToggle.setAttribute(
        "aria-label",
        isDark
            ? "Switch to light theme"
            : "Switch to dark theme"
    );

    themeIcon.textContent = isDark ? "🌙" : "☀️";
    themeText.textContent = isDark
        ? "Dark mode"
        : "Light mode";
}

const initialTheme =
    savedTheme === "light"
        ? "light"
        : "dark";

applyTheme(initialTheme);
```



• **CONTRACT-FIRST CONSTRAINTS:**

- **• State persistence strictly via localStorage key 'theme'.**

```
const initialTheme =
    savedTheme === "light"
        ? "light"
        : "dark";

applyTheme(initialTheme);

themeToggle.addEventListener("click", () => {

    const currentTheme =
        document.documentElement.dataset.theme;

    const nextTheme =
        currentTheme === "dark"
            ? "light"
            : "dark";

    localStorage.setItem("theme", nextTheme);

    applyTheme(nextTheme);
```

- **• Colors paired via CSS variables; zero hardcoded hex codes in rules.**

```
body {
    margin: 0;
    min-width: 0;
    font-family: var(--font-family);
    background: var(--color-bg);
    color: var(--color-text);
    line-height: 1.6;
}
```

- **• Performance Budget: Zero CLS, LCP < 2.0s on DevTools Fast 3G.**

```css
img {
    display: block;
    max-width: 100%;
    height: auto;
}
```

- **Hero Section: High-res portrait with explicit dimensions, headline, pitch.**

```
</section>

    <!-- HERO -->
    <section class="hero container" aria-labelledby="hero-title">

        <div class="hero-content">

            <p class="eyebrow">HELLO, I'M CHINH</p>

            <h2 id="hero-title">
                Building scalable web and AI solutions.
            </h2>

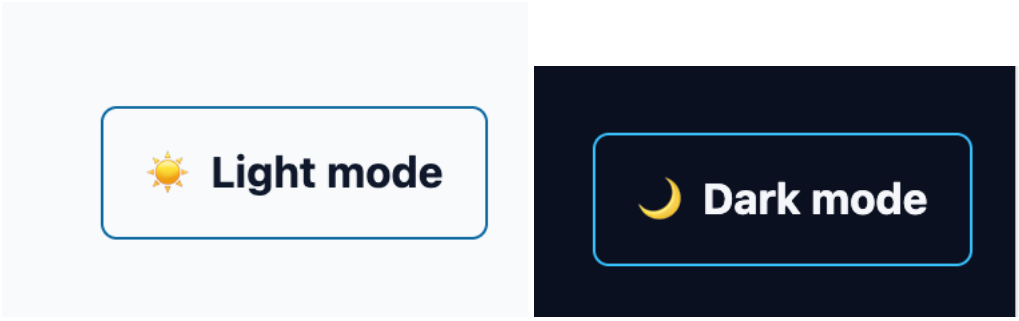
            <p>
                I develop modern web applications and machine learning
                systems with a focus on clean architecture,
                accessibility and maintainable code.
            </p>

            <a
                class="button"
                href="#projects"
            >
                View my projects
            </a>

        </div>

        <div class="hero-placeholder" aria-label="Developer portrait">
            <span aria-hidden="true">👨‍💻</span>
        </div>
```

- **Theme Switcher: Accessible button with aria-pressed and dynamic icon.**



- **Skills Matrix: Categorized badges arranged in clean CSS Grid.**

```css
.skills-grid {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: var(--spacing-lg);
}
```

- **Project Cards: Self-contained <article> blocks with tags, links, and descriptions.**

```css
.project-card {
    display: flex;
    flex-direction: column;

    min-width: 0;

    padding: var(--spacing-lg);

    background: var(--color-surface);
    border: 1px solid var(--color-border);
    border-radius: var(--radius-md);

    box-shadow: var(--shadow-card);
}
```

- **Contact Form: Validated native form with client-side state handling.**

```
<!-- CONTACT -->
<section
    class="section container"
    id="contact"
    aria-labelledby="contact-title"
>

    <div class="section-heading">

        <p class="eyebrow">GET IN TOUCH</p>

        <h2 id="contact-title">
            Contact Me
        </h2>

    </div>


    <form
        class="contact-form"
        id="contact-form"
    >

        <div class="form-group">

            <label for="name">
                Name
            </label>

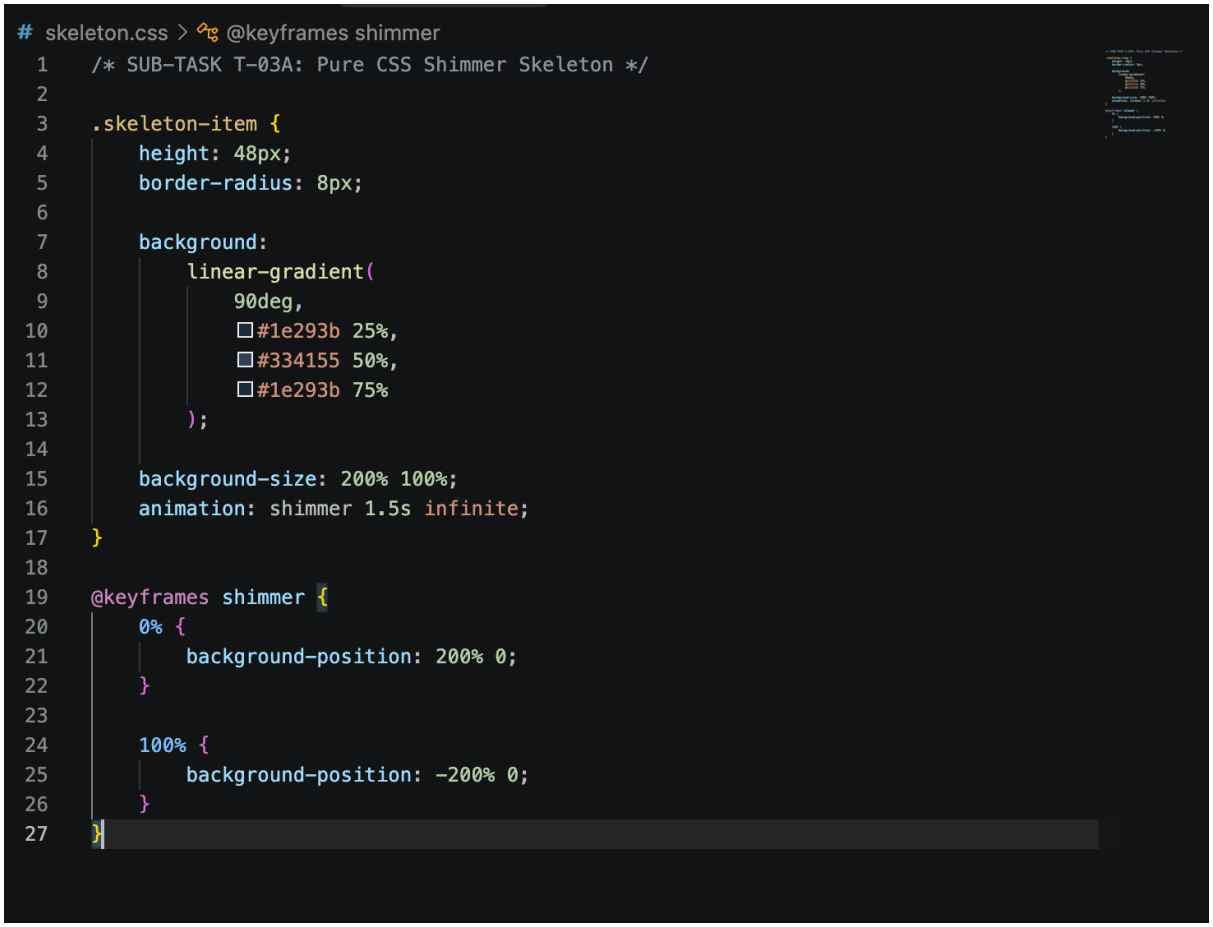
            <input
                id="name"
                name="name"
                type="text"
```

EXERCISE 3

Link AI prompt: https://chatgpt.com/share/6ab6a365-054c-83ec-8d53-4b20153ccef3?ogimg=plain

Screenshot for results

- **SUB-TASK T-03A: Loading Skeleton (Pure CSS Shimmer gradient).**



- **SUB-TASK T-03B: Live Data State (Flexbox metadata badges & Grid list).**

```
<section
    id="live-state"
    class="component-state"
    aria-labelledby="live-title"
>

    <div class="container">

        <h2 id="live-title">Project Data</h2>

        <div class="metadata-badges">
            <span class="metadata-badge">
                Status: Active
            </span>

            <span class="metadata-badge">
                Projects: 6
            </span>

            <span class="metadata-badge">
                Updated: Today
            </span>
        </div>

        <div class="data-grid">

            <article class="data-card">
                <h3>Portfolio Website</h3>
                <p>HTML, CSS, JavaScript</p>
            </article>
```

• **SUB-TASK T-03C: Empty & Error States with accessible retry trigger.**

```
<section
    id="empty-state"
    class="component-state"
    aria-labelledby="empty-title"
>

    <div class="container empty-state">

        <h2 id="empty-title">
            No data available
        </h2>

        <p>
            There is currently no project data to display.
        </p>

        <button
            type="button"
            id="empty-retry"
        >
            Try Again
        </button>

    </div>

</section>
```

- **DECOMPOSITION SUBMISSION PIPELINE:**
- **• Define state machine in TASK_DECOMPOSITION.md before prompting AI.**

Exercise 3 - Resilient Component Architecture

State Machine

The resilient component has four states:

1. LOADING
   - Display CSS shimmer skeleton.
   - Hide live data.
   - No JavaScript-generated skeleton.

2. LIVE
   - Display metadata badges.
   - Display data items using CSS Grid.
   - Hide loading, empty and error states.

3. EMPTY
   - Display an accessible empty-state message.
   - Provide a clear action to retry/load data.

4. ERROR
   - Display an error message.
   - Provide an accessible Retry button.
   - Retry returns the component to LOADING state.